Ollivier–Ricci Curvature as a Landscape Analysis Framework for Discrete Combinatorial Optimization

Roberto Sergiu Hordoan[0009−0004−5150−6594]

Babeş–Bolyai University, Cluj-Napoca, Romania
roberto.hordoan@stud.ubbcluj.ro

Abstract. We introduce the *ORC Transition Graph* (OTG), a deterministic directed graph over local optima derived from Ollivier–Ricci curvature (ORC), as a structural analysis tool for discrete fitness landscapes. Each local optimum has exactly one outgoing edge to the optimum reached by following its most-negative-curvature direction and hill-climbing—a purely local, $O(k^2)$ computation. We study OTG on NK landscapes, epistatic W-model benchmarks ($N = 16$, full enumeration), random MAX-SAT (N up to 100), QAP (permutation swaps, n up to 9), and TSP (2-opt, n up to 50 via sampling). The OTG extracts cross-basin information invisible to standard local optima networks (LONs): using only 1-hop neighborhood data, it achieves up to 9.2× better attractor quality than LON-d1 at equal information cost on epistatic W-model landscapes, and even outperforms the stochastic 3-hop LON-d3 on those instances. A structural crossover in OTG topology—from acyclic to cyclic at moderate epistasis—correlates with when ORC-guided perturbation outperforms random perturbation in iterated local search (ILS). The OTG framework applies to any graph with the disjoint-neighborhood property (Proposition 1), which we prove covers binary hypercubes, permutation swap graphs, *and* TSP 2-opt neighborhoods—the three foundational neighborhoods in combinatorial optimization. ORC's per-step escape advantage grows monotonically with neighborhood size—from 3.1× to 18.3× over random on TSP ($n = 8$ to 50)—though its computational cost ($O(k^2)$ per optimum) limits algorithmic use to settings where fitness evaluations are expensive. Per-optimum ORC metrics converge from ∼20 sampled optima (<1% error), while full OTG structural analysis remains limited to enumerable landscapes.

Keywords: Fitness landscape analysis · Ollivier–Ricci curvature · Local optima networks · Discrete optimization · Permutation spaces

1 Introduction

Fitness landscape analysis (FLA) provides the theoretical foundation for understanding why metaheuristic optimizers succeed or fail on specific problem

instances [16] and for automated algorithm selection [4]. Classical metrics—fitness-distance correlation (FDC) [2], autocorrelation length [19], information content [23], and exploratory landscape analysis (ELA) features [8]—characterize global statistical properties such as ruggedness and smoothness. Local optima networks (LONs) [12,14] go further by representing basin connectivity as a graph whose edges are discovered through stochastic perturbation followed by hill climbing (HC).

Both approaches share a limitation: classical metrics provide scalar summaries blind to local geometric structure, while LONs provide connectivity but not *directional* information—at a local optimum, which specific neighbor escapes the basin? LON construction also depends on perturbation distance d and requires many stochastic trials per optimum.

We address these limitations through Ollivier–Ricci curvature (ORC) [15], a discrete analogue of Ricci curvature that measures how neighborhood distributions of adjacent nodes diverge or converge. Negative curvature identifies directions where the landscape geometry "spreads apart"—the signature of inter-basin saddle regions. From ORC, we construct the *ORC Transition Graph* (OTG): a deterministic directed graph where each local optimum points to the optimum reached by following its min-ORC direction.

Contributions.

1. We formalize fitness-lifted ORC for discrete optimization landscapes and prove that negative curvature is guaranteed at every local optimum on graphs with the disjoint-neighborhood property, covering binary hypercubes, permutation swap graphs, and TSP 2-opt neighborhoods.
2. We introduce the OTG, a deterministic, parameter-free directed graph over local optima derived from optimal-transport-based curvature. On epistatic binary landscapes, the OTG achieves up to 9.2× better attractor quality than LON-d1 at equal information cost; across binary, permutation, and tour search spaces, ORC's per-step escape advantage grows monotonically with neighborhood size.
3. We validate the framework algorithmically through ORC-guided iterated local search (ILS) and show that OTG structural properties—specifically the presence or absence of cycles—indicate when ORC-guided perturbation adds value over random perturbation.

2 Background

A *fitness landscape* $(S, \mathcal{N}, f)$ consists of solutions S, neighborhood function $\mathcal{N} : S \to 2^S$, and fitness $f : S \to \mathbb{R}$ [19]. A solution x^* is a *local optimum* (minimization) if $f(x^*) \leq f(y)$ for all $y \in \mathcal{N}(x^*)$. LONs [12,13] compress the landscape into a graph whose nodes are local optima and whose edges encode transitions discovered by perturbing a solution (flipping d random bits) and hill-climbing. LON features—including funnel structure [11], fractal dimension [20],

and phase-transition indicators [10]—predict search difficulty [14], but edge construction requires T stochastic trials per optimum (typically $T \geq 50$) and the resulting funnel decomposition depends on d.

Ollivier [15] defined Ricci curvature for metric-measure spaces; Lin et al. [6] gave a modified graph formulation. For a graph edge (x, y):

$$\kappa(x, y) = 1 - \frac{W_1(\mu_x, \mu_y)}{d(x, y)} \tag{1}$$

where μ_x, μ_y are neighborhood probability measures and W_1 is the Wasserstein-1 distance [24]. Negative curvature signals diverging neighborhoods—a signature of community bridges [9] and GNN bottlenecks [22,18]; positive curvature signals contraction. We choose Ollivier's formulation over Forman's [17]: on triangle-free graphs (including all three neighborhoods in Corollary 1), Forman curvature is the constant $4-2k$ for every edge on unweighted k-regular graphs, and depends only on endpoint-local quantities even with fitness weighting—missing the distributional matching that gives ORC its cross-basin signal. To our knowledge, ORC has not been applied to fitness landscape analysis.

3 ORC on Fitness Landscapes

3.1 Fitness-Lifted ORC

We define ORC on the search graph $(S, \mathcal{N})$ with uniform measures μ_x over $\{x\} \cup \mathcal{N}(x)$. The transport cost between $a \in \text{supp}(\mu_x)$ and $b \in \text{supp}(\mu_y)$ is:

$$c(a, b) = \delta(a, b) + \gamma \cdot |f(a) - f(b)| \tag{2}$$

where $\gamma \geq 0$ controls fitness sensitivity and the overlap indicator is:

$$\delta(a, b) = \begin{cases} 0 & \text{if } a = b \\ 1 & \text{if } a \text{ or } b \in \text{supp}(\mu_x) \cap \text{supp}(\mu_y) \\ 2 & \text{otherwise (both exclusive)} \end{cases} \tag{3}$$

This is a conservative coarse-graining of graph distance computable from 2-hop information alone, preserving $O(k^2)$ locality. For uniform measures of equal support size, W_1 reduces to an optimal assignment problem solved by the Hungarian algorithm in $O(k^3)$ [5]; the fine-grained complexity of this reduction has been formally established [1].

Parameter sensitivity. We use $\gamma = 1$ throughout. Importantly, the ORC-guided *direction* (Algorithm 1) is γ-invariant: changing γ scales all fitness differences uniformly, so the optimal transport matching order—and thus the min-ORC neighbor—does not change. Empirically, escape rates are identical to three decimal places across $\gamma \in \{0.1, 0.5, 1.0, 2.0\}$ on all 480 instances; only absolute curvature values change. The framework is thus parameter-free for navigation; γ matters only for difficulty prediction, where $\gamma = 1$ balances structural and fitness terms. This invariance is a consequence of the *linear* fitness term in Eq. (2): a nonlinear cost $g(|f(a) - f(b)|)$ would break it, though such extensions are a natural direction for future work.

3.2 Why ORC Sees What 1-Hop Methods Cannot

Proposition 1 (ORC Cross-Neighborhood Decomposition). *Let $G = (V, E)$ be a regular graph of degree k in which adjacent nodes share no common neighbors: $\mathcal{N}(x) \cap \mathcal{N}(y) = \emptyset$ for all $(x, y) \in E$. Then $|\mathrm{supp}(\mu_{x^*}) \cap \mathrm{supp}(\mu_y)| = 2$ (only x^* and y are shared), and the optimal transport cost decomposes as:*

$$(k{+}1)\, W_1(\mu_{x^*}, \mu_y) = C_{\text{shared}} + \min_{\sigma} \sum_{i=1}^{k-1} \big[2 + \gamma\, |f(a_i) - f(b_{\sigma(i)})|\big] \tag{4}$$

where $C_{\text{shared}} \leq 2 + \gamma(\Delta + \Delta')$ accounts for the two shared elements, $\{a_i\} = \mathcal{N}(x^*) \setminus \{y\}$, $\{b_j\} = \mathcal{N}(y) \setminus \{x^*\}$ are the $k{-}1$ exclusive neighbors on each side, and σ ranges over permutations of $\{1, \ldots, k{-}1\}$.

Proof. Since $\mathcal{N}(x^*) \cap \mathcal{N}(y) = \emptyset$, the only shared support elements are x^* and y themselves. Any assignment $\pi : \mathrm{supp}(\mu_{x^*}) \to \mathrm{supp}(\mu_y)$ partitions into: (i) the two shared positions (x^*, y) contributing C_{shared}, and (ii) the k–1 exclusive pairs, each incurring $\delta = 2$ plus a fitness-difference term. The optimal σ minimizes the total fitness-comparison cost.

Corollary 1 (Applicability to Standard Search Spaces). *Proposition 1 applies to: **(i)** the N-dimensional binary hypercube with bit-flip neighborhood ($k = N$), **(ii)** the symmetric group S_n with pairwise-swap neighborhood ($k = \binom{n}{2}$), and **(iii)** Hamiltonian tours with 2-opt neighborhood ($k = n(n{-}3)/2$).*

Proof. **(i)** On the hypercube, flipping bit i in x and bit j in $y = x \oplus e_p$ yields the same neighbor only if $e_i = e_p \oplus e_j$; since single bit-flips cannot equal a two-bit change, $\mathcal{N}(x) \cap \mathcal{N}(y) = \emptyset$. **(ii)** On S_n, if $y = x \circ \tau_{ab}$, a common neighbor z requires $\tau_{ij} = \tau_{ab} \circ \tau_{kl}$ for transpositions τ_{ij}, τ_{kl}. The product of two distinct transpositions is a 3-cycle or double transposition—never a transposition. **(iii)** A 2-opt move is a segment reversal $\rho_{p,q}$ mapping position $i \mapsto p{+}q{+}1{-}i$ for $i \in \{p{+}1, \ldots, q\}$ and fixing all others—an affine map with *slope* -1 on its support. A common neighbor requires $\rho_{a,b} = \rho_{c,d} \circ \rho_{p,q}$ for distinct non-trivial reversals. *Disjoint supports:* the composition has non-contiguous support (two separated intervals with a fixed gap), impossible for a single reversal. *One nested in the other* (WLOG $\mathrm{supp}(\rho_{p,q}) \subset \mathrm{supp}(\rho_{c,d})$): on the inner interval, $(\rho_{c,d} \circ \rho_{p,q})(i) = i + \delta$ is a *translation* (slope $+1$), since each reversal contributes slope -1 and $(-1)(-1) = +1$. Any single reversal has slope -1 on its entire support, so the composition cannot be one. *Partial overlap:* within the overlap the composition transitions from slope $+1$ (where both act) to slope -1 (where only one acts)—again incompatible with a single reversal. Verified computationally at $n \leq 8$ (all 25,200 edges) and by sampling at $n \leq 11$ (all >200,000 tested edges have overlap 0).

These three search spaces—binary strings (SAT, NK), permutations with swaps (QAP, scheduling), and tours with 2-opt (TSP)—cover the foundational neighborhoods in combinatorial optimization.

Algorithm 1 ORC-Guided Escape from a Local Optimum

Require: Local optimum x^*, fitness f, neighborhood $\mathcal{N}$, parameter γ
1: **for** each $y \in \mathcal{N}(x^*)$ **do**
2: Compute $\kappa(x^*, y)$ via Eq. (1)–(3)
3: **end for**
4: $y^* \leftarrow \arg\min_y \kappa(x^*, y)$
5: **return** HillClimb(y^*)

The exclusive-pair term is the key: it computes an optimal matching between the fitness distributions visible from x^* and from y. At a basin boundary, the two sides' fitness distributions diverge, making curvature strongly negative. By contrast, LON-d1 evaluates each neighbor y using only $f(y)$ and the downstream hill-climbing destination, discarding the $(k-1)$ exclusive neighbors entirely—precisely the distributional comparison that ORC captures.

3.3 Negative Curvature Guarantee

Theorem 1. *Let x^* be a strict local optimum on any degree-k graph satisfying the disjoint-neighborhood condition of Proposition 1 ($k \geq 4$), with cost* (2)–(3) *and $\gamma > 0$. If neighbor y satisfies $\Delta = f(y) - f(x^*) < (k-3)/((k+1)\gamma)$, then $\kappa(x^*, y) < 0$.*

Proof. From Proposition 1, every assignment incurs exclusive-pair structural cost at least $2(k-1)$, giving $W_1 \geq 2(k-1)/(k+1)$. With edge cost $c(x^*, y) = 1 + \gamma\Delta$: $\kappa(x^*, y) \leq 1 - 2(k-1)/[(k+1)(1+\gamma\Delta)]$, which is negative iff $\Delta < (k-3)/((k+1)\gamma)$.

For the binary hypercube with $k = N = 16$, $\gamma = 1$: the bound requires $\Delta < 0.76$. For QAP with $n = 8$ ($k = 28$): $\Delta < 0.86$. Across all 5,000+ local optima in our binary benchmarks and 1,500+ QAP optima, **100% satisfy the condition**. The min-ORC neighbor $y^* = \arg\min_y \kappa(x^*, y)$ identifies the direction of maximal neighborhood divergence—the strongest saddle direction (Alg. 1).

3.4 ORC-Guided Escape and the OTG

Applying Algorithm 1 at every local optimum of a landscape defines the *ORC Transition Graph* (OTG): a deterministic directed graph where each optimum has exactly one outgoing edge to the optimum reached by following its min-ORC neighbor and hill-climbing. The OTG is a functional graph; its terminal attractors are either self-loops (sinks) or directed cycles. The *funnel* of an attractor is the set of all optima whose OTG path converges to it. Unlike LONs [21], the OTG is deterministic and parameter-free: it requires no perturbation distance or stochastic trials.

4 Experimental Setup

4.1 Benchmark Problems

We evaluate on five benchmark families spanning three search spaces:

NK landscapes [3] define tunably rugged fitness via K epistatic interactions per bit on length-N binary strings. We test $K \in \{4, 8, 12\}$ at $N = 16$ (30 instances each, full enumeration).

W-model [25] applies neutrality, epistasis, and ruggedness transformations controlled by a difficulty parameter ν to a base OneMax problem. We test $\nu \in \{3, 4, 6\}$ at $N = 16$ (30 instances each, full enumeration).

Random MAX-SAT. Random 3-SAT instances with $N \in \{20, 50, 100\}$ variables and clause-to-variable ratio $\alpha \in \{3.0, \ldots, 6.0\}$ (20 instances each). Local optima are collected via random-restart HC (500–2,000 restarts). This domain has well-studied phase transitions [10] and no structural overlap with NK or W-model.

Random QAP. Quadratic assignment problems with uniform random flow/distance matrices and pairwise-swap neighborhood, $n \in \{6, 7, 8, 9\}$ (20 instances each, full enumeration).

Random TSP. Random-matrix TSP instances with 2-opt neighborhood. Full enumeration at $n \in \{8, 9, 10\}$ (20 each, 4–7 optima per instance); sampling-based analysis at $n \in \{15, 20, 30, 50\}$ (20 each, 300 random-restart 2-opt runs per instance). TSP instances at small n serve to validate the theoretical framework on an exact ground truth; sampling-based runs at larger n test scaling.

The five families were chosen to cover three distinct search spaces (binary strings, permutations, tours), a range of landscape ruggedness levels, and both enumerable and sampling-only regimes.

4.2 Algorithms and Baselines

Hill Climbing (HC). Best-improvement (steepest descent) HC: at each step, all k neighbors are evaluated and the one with the best fitness is selected. HC terminates when no neighbor improves on the current solution, yielding a local optimum. Ties are broken uniformly at random.

Random-Restart Hill Climbing (RR-HC). Repeat HC from uniformly random initial solutions until the budget of fitness evaluations (FEs) is exhausted. Return the best local optimum found.

Iterated Local Search (ILS) [7]. From a local optimum, apply a perturbation (bit-flips) and re-run HC; repeat until the FE budget is exhausted. We test four ILS variants, all using a fixed perturbation strength of 3 bit-flips for fair comparison:

- **Rand-ILS**: flip 3 random bits.
- **ORC+Pert**: flip the min-ORC bit (Algorithm 1) plus 2 random bits.
- **ORC-ILS**: flip only the min-ORC bit (1 bit; no random component).
- **MG-ILS**: flip only the MinGap bit (1 bit; see below).

Table 1. ORC escape direction analysis on binary landscapes ($N = 16$, full enumeration). %ORC: fraction of local optima where Algorithm 1 reaches a better optimum. %MinGap: same using the argmin-$|\Delta|$ heuristic. %Rand: random neighbor baseline (30 trials). The ORC advantage over MinGap emerges on epistatic W-model landscapes where the fitness-distance relationship is distorted.

Config	#Opt	%ORC	%MinGap	%Rand	ORC/R	ORC/MG	Agree
NK $K=4$	170	59.6	57.4	31.6	1.9×	1.04×	89.6%
NK $K=8$	888	58.5	56.5	33.6	1.7×	1.04×	90.3%
NK $K=12$	2249	51.1	49.4	37.6	1.4×	1.03×	91.7%
W $\nu=3$	406	79.9	43.5	29.4	**2.7**×	**1.84**×	59.6%
W $\nu=4$	3732	59.7	25.9	39.5	1.5×	**2.31**×	43.3%
W $\nu=6$	5836	63.9	37.4	48.6	1.3×	1.71×	52.7%

ORC-ILS and MG-ILS use only 1-bit perturbation; they are included as ablations to isolate the effect of the directed bit.

MinGap. The neighbor y with the smallest fitness difference $|f(y) - f(x^*)|$ (argmin-$|\Delta|$ heuristic). This greedy baseline uses only fitness values, not distributional information.

LON-d1. At a local optimum x^*, the LON-d1 destination is the mode HC outcome over all k single-bit-flip neighbors (1-hop, deterministic).

LON-d3. At a local optimum x^*, the LON-d3 destination is the mode HC outcome from 100 random 3-bit perturbations (3-hop, stochastic).

4.3 Evaluation Metrics

Escape rate (%). Fraction of local optima where a method reaches a strictly better optimum after one perturbation + HC step. This is the primary metric shared across all experiments.

Mean terminal rank. Normalized fitness rank of the attractor reached by following OTG/LON paths (0.0 = best, 0.5 = random).

Top-5% attractor (%). Fraction of optima whose OTG/LON path terminates at an attractor in the top 5% by fitness.

Compression. #funnels / #optima—lower means better funnel aggregation.

Success rate (%). Fraction of ILS runs finding the global optimum within the FE budget (used for the ILS comparison in Section 5.4).

5 Results

5.1 Single-Step Escape Analysis

Table 1 presents the single-step escape analysis. On NK landscapes, ORC beats random by 1.4–1.9× but only marginally outperforms MinGap (1.03–1.04×); on

Table 2. OTG vs. LON funnel quality ($N = 16$, full enumeration). LON-d1: 1-hop, deterministic. LON-d3: 3-hop, 100 stochastic trials. Rank: 0.0 = best, 0.5 = random. Compression: #funnels / #optima.

		Mean terminal rank			Top-5% attractor			Compression		
Config	**#Opt**	**OTG**	**d1**	**d3**	**OTG**	**d1**	**d3**	**OTG**	**d1**	**d3**
NK $K=4$	170	.255	.490	**.181**	24.6	5.6	**30.1**	40	98	35
NK $K=8$	888	.231	.476	**.099**	24.0	6.0	**46.1**	42	97	16
W $\nu=3$	406	**.037**	.340	.077	**90.7**	23.2	77.3	15	98	21
W $\nu=4$	3732	.073	.258	**.012**	73.7	20.0	**97.8**	13	75	2
W $\nu=6$	5836	.066	.221	**.011**	68.5	18.3	**96.3**	14	68	1

these regular Hamming graphs the fitness gradient already captures most of the cross-basin signal. On W-model landscapes, where epistasis distorts the fitness-distance relationship, ORC's advantage over MinGap becomes substantial: 2.31× at $\nu=4$, with only 43% agreement between the two methods.

When ORC and MinGap disagree, ORC wins. On W $\nu = 3$, ORC's choice leads to a better basin in **78.7%** of disagreements vs. MinGap's 0.3% (21% ties). On NK $K=4,8$, ORC wins 43–44% vs. MinGap's 24–25%. MinGap almost never outperforms ORC when the two disagree, confirming that ORC's optimal transport matching captures cross-basin structure that the fitness-gap heuristic misses.

ORC ranks all escape directions. Spearman correlation between ORC values and destination quality: mean $\rho=0.44$ on W $\nu=3$ (95% positive), $\rho=0.33$ on NK—more negative curvature $\rightarrow$ better destination basin.

5.2 OTG vs. LON: Funnel Detection from Local Data

Table 2 compares OTG with LON variants. Both OTG and LON-d1 use only 1-hop data, making this an equal-information comparison. LON-d1 yields 97–98% singleton funnels; OTG compresses to 13–42% funnels with substantially better attractor quality. On W $\nu=3$: OTG rank 0.037 vs. LON-d1's 0.340 (9.2× improvement); 90.7% reach top-5% vs. 23.2%. The OTG even outperforms LON-d3 (3-hop, stochastic) on W $\nu = 3$ (rank 0.037 vs. 0.077); at higher epistasis ($\nu \geq 4$), LON-d3's wider perturbation provides superior attractor quality. Path to top-5% attractors: median 2 hops, P90 3 hops.

5.3 Scaling Across Problem Sizes and Search Spaces

MAX-SAT (binary strings). Table 3 summarizes results. Proposition 1's sort-and-match reduction lowers per-edge ORC cost from $O(k^3)$ to $O(k \log k)$, enabling computation at 500 optima for $N = 100$ in ∼18 s. The ORC/Random escape ratio increases from 2.4× at $N = 20$ to 3.7× at $N = 100$ ($\alpha = 4.27$; $\rho(N, \text{ORC/Rand}) = 0.77$, $p < 10^{-84}$). As N grows, the (N–1) exclusive-neighbor

Table 3. ORC escape advantage on random MAX-SAT. The ORC/Random ratio *increases* with N: ORC's distributional signal becomes more valuable in higher dimensions. 20 instances per cell; $\alpha_c \approx 4.27$ is the satisfiability phase transition.

N	α	%ORC	%Rand	%MG	ORC/R	ORC/MG	Mean ORC
20	3.00	34.3	13.5	26.1	2.5×	1.3×	−0.428
20	4.27	42.7	17.5	36.7	2.4×	1.2×	−0.374
20	6.00	50.0	20.1	46.3	2.5×	1.1×	−0.320
50	3.00	32.8	10.1	21.9	3.3×	1.5×	−0.493
50	4.27	37.3	10.6	31.3	3.5×	1.2×	−0.418
50	6.00	40.4	10.3	38.1	3.9×	1.1×	−0.353
100	3.00	34.0	9.9	21.4	3.4×	1.6×	−0.524
100	4.27	39.5	10.7	32.0	3.7×	1.2×	−0.450
100	6.00	42.7	10.1	40.2	4.2×	1.1×	−0.385

Table 4. OTG vs. LON-d1 on QAP (swap neighborhood, full enumeration). ORC's advantage grows with n. LON-d1 degrades to 100% singletons at $n = 9$. 20 instances per row.

				Escape rate		Mean rank		Compression	
n	k	#Opt	ORC/R	ORC	LON	OTG	LON	OTG	LON
6	15	12	1.8×	51.2	4.3	.230	.466	49%	96%
7	21	27	2.1×	55.9	3.0	.235	.476	44%	97%
8	28	78	2.3×	58.7	0.9	.230	.493	41%	99%
9	36	233	**2.7**×	65.0	0.1	**.194**	.499	35%	**100%**

comparisons carry more distributional information, strengthening the cross-basin signal. Mean ORC also tracks the SAT phase transition: $\rho(\text{ORC}, \alpha) = 0.94$ ($N = 20$), 0.99 ($N = 100$).

QAP (permutations). Table 4 shows that LON-d1 degrades to 0.1% escape at $n = 9$ (100% singletons, rank $\approx$ random); OTG improves: ORC/Random 1.8× → 2.7×, rank 0.194 with 35% compression. ORC's advantage over MinGap is modest (1.06×) on these random instances; the primary result is OTG vs. LON-d1.

TSP (2-opt). Table 5 confirms Corollary 1(iii): the disjoint-neighborhood property holds at all tested sizes. As k grows from 20 to 1175, random escape drops from 25% to 2.5% while ORC maintains 36–45%, yielding an ORC/Random ratio that grows monotonically ($\rho(k, \text{ORC/Rand}) > 0.99$). ORC also consistently outperforms MinGap, winning 45–52% of disagreements vs. 9–22%. This pattern—growing ORC/Random advantage, stable ORC escape rate—is consistent across all three search spaces, confirming that ORC's distributional matching becomes more valuable as neighborhood size increases.

Table 5. ORC escape on TSP 2-opt across scales. Top rows: full enumeration. Bottom rows: sampling-based (300 restarts per instance). The ORC/Random ratio grows monotonically from 3.1× at $n=8$ to 18.3× at $n=50$. When ORC and MinGap disagree, ORC wins 45–52% vs. MinGap's 9–22%.

n	k	#Opt	Escape rate (%) ORC	MG	Rand	ORC/R	ORC/MG
Full enumeration							
8	20	4	77.5	66.7	25.0	3.1×	1.16×
9	27	5	64.7	57.5	21.1	3.1×	1.13×
10	35	7	68.9	62.6	21.3	3.2×	1.10×
Sampling-based (300 restarts)							
15	90	225	40.0	35.4	5.2	7.7×	1.13×
20	170	289	35.7	32.9	3.9	9.2×	1.09×
30	405	300	40.7	38.7	3.2	12.7×	1.05×
50	1175	300	45.0	43.5	2.5	18.3×	1.03×

Table 6. Fixed-budget ILS comparison at $N=16$ (5,000 FEs, 30 instances × 30 trials). ORC+Pert and Rand-ILS both flip 3 bits per perturbation; ORC-ILS and MG-ILS flip only 1 directed bit (ablations). Bold: best per row.

Config	ORC+Pert	Rand-ILS	RR-HC	ORC-ILS	MG-ILS
Success rate (% finding global optimum)					
NK $K=4$	77.3	**88.9**	88.4	6.7	6.1
NK $K=8$	42.0	**44.9**	45.2	1.3	1.3
W $\nu=3$	46.7	**80.2**	52.2	8.4	1.8
W $\nu=4$	**89.9**	69.3	39.7	1.6	1.0
W $\nu=6$	**87.3**	56.0	26.3	1.6	0.9

5.4 ORC-Guided Iterated Local Search

Table 6 compares ILS variants under a fixed 5,000 FE budget. ORC+Pert wins on 2 of 5 configurations and loses on 3, partitioning cleanly along OTG topology. In the *acyclic regime* (NK, W $\nu=3$), Rand-ILS dominates—sometimes by large margins (80.2% vs. 46.7% on W $\nu=3$)—because the landscape has clear funnel structure and ORC's directional constraint reduces exploration diversity. In the *cyclic regime* (W $\nu = 4,6$), ORC+Pert dominates: 89.9% vs. 69.3% at $\nu = 4$. ORC-ILS and MG-ILS, which flip only a single directed bit, perform poorly (1–8% success) because without random bits their perturbation deterministically revisits the same path and becomes trapped in OTG cycles.

Computational cost. At $N = 16$, ORC costs ∼120 FEs (∼3 ms); at $N = 100$, ∼5 000 FEs (∼18 ms). At $N = 100$ on MAX-SAT (10 s wall-clock budget), Rand-ILS completes 25× more iterations and finds better solutions; ORC's 3.7× per-step advantage cannot overcome the iteration deficit when evaluations cost

Table 7. Spearman correlations (ρ) between landscape metrics and algorithm performance ($N = 16$, 480 instances: 30 each for NK $K \in \{4, 8, 12, 15\}$ and W-model $\nu \in \{1, \ldots, 12\}$). ***: $p < 0.001$. Best per column in **bold**. †: requires full enumeration. HC: best-improvement hill climbing (success = global optimum found). EA: (μ+λ) evolutionary algorithm with μ=5, λ=10, bit-flip mutation (success rate). RS: random sampling baseline (mean fitness of 10^4 uniform samples).

Feature	Cost	HC succ.	EA succ.	RS mean
Mean ORC (ours)	$O(Rk^2)$	-0.670^{***}	$\mathbf{-0.652}^{***}$	-0.455^{***}
%ORC→better	$O(Rk^2)$	-0.435^{***}	-0.174^{***}	-0.615^{***}
FDC	$O(2^N)$	0.276^{***}	0.501^{***}	0.024
Autocorrelation	$O(L)$	0.543^{***}	0.544^{***}	0.372^{***}
Info. content H	$O(L)$	-0.308^{***}	0.306^{***}	-0.585^{***}
Mean $\lvert\Delta f\rvert$ (ELA)	$O(L)$	$\mathbf{-0.695}^{***}$	-0.083	$\mathbf{-0.769}^{***}$
NBC mean (ELA)	$O(L)$	-0.419^{***}	0.072	-0.487^{***}
Basin entropy†	$O(2^N)$	-0.731^{***}	-0.460^{***}	-0.610^{***}
OTG attractor rank†	$O(2^N)$	0.636^{***}	0.157	0.754^{***}
OTG compression†	$O(2^N)$	0.589^{***}	-0.094	0.748^{***}

$\sim$0.001 ms. ORC-guided search becomes cost-effective when individual evaluations cost ≥ 1 ms (e.g., simulation-based optimization).

6 Discussion

6.1 OTG Structural Properties

Figure 1 reveals a structural crossover: at W $\nu = 3$ the OTG is acyclic (91% top-5%); at ν=4, cycles emerge trapping 21% of optima (top-5% drops to 74%). NK landscapes show no cycles at any K. This crossover correlates with the ILS results in Table 6: in acyclic regimes, random perturbation suffices and ORC's directional constraint reduces diversity; in cyclic regimes, ORC+Pert's directed bit breaks cycles that trap random perturbation. We note this is an observed correlation on the tested benchmark families; validating it as a predictive criterion on unseen landscape types is an avenue for future work.

6.2 ORC as Difficulty Predictor

Table 7 evaluates ORC as a difficulty predictor alongside classical FLA metrics. Mean $|\Delta f|$ achieves the strongest individual correlation on HC ($\rho = -0.695$) and RS ($\rho = -0.769$), but fails on EA ($\rho = -0.083$, not significant). Information content and NBC show similar target-specificity, with sign reversals across algorithm types. Autocorrelation maintains consistent sign but with weaker magnitude on RS ($\rho = 0.372$). Mean ORC is the only tractable metric with strong correlations (all $|\rho| > 0.45$) of consistent sign across all three algorithm types, making it a

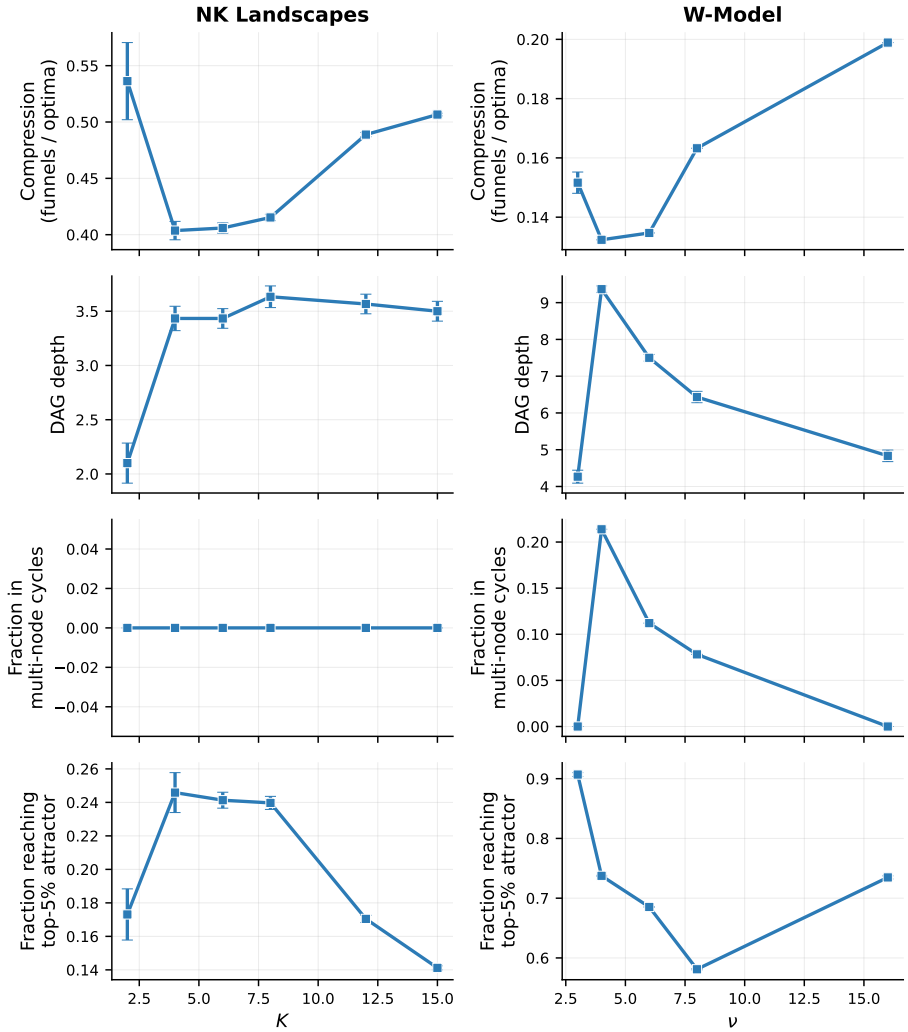

Fig. 1. OTG structural properties vs. ruggedness parameter. Left column: NK landscapes (K). Right column: W-model (ν). A structural crossover in cycle structure appears at W $\nu = 4$: multi-node cycles emerge (21% of optima), DAG depth peaks, and top-5% reachability drops. NK landscapes show no cycles at any K.

robust general-purpose predictor that does not require choosing the right metric for a specific algorithm family.

Table 8 extends the escape analysis to $N = 20$, confirming the pattern beyond the $N = 16$ regime: ORC's advantage decreases monotonically with K/N as maximal-ruggedness landscapes leave no exploitable distributional structure.

Table 8. ORC escape at $N=20$ (NK landscapes, full enumeration, 10 instances each). The ORC/Random advantage replicates the $N = 16$ pattern; at $K = 19$ ($\approx N$), all neighbors are equally rugged and ORC's advantage vanishes.

K	#Opt	%ORC	%Rand	ORC/R
4	557	62.3	31.2	2.0×
8	5,012	62.5	34.1	1.8×
15	26,086	52.9	37.2	1.4×
19	49,966	49.5	47.5	1.0×

6.3 OTG Visualization

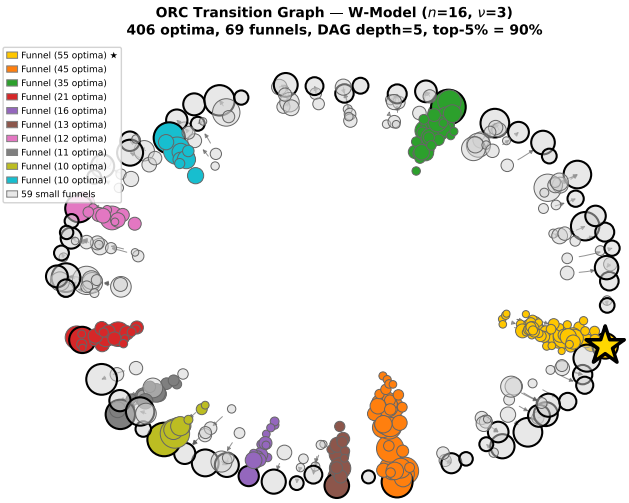

Fig. 2. OTG for a W-model instance ($N = 16$, $\nu = 3$, seed 0). Each node is a local optimum, sized by basin of attraction and colored by funnel membership. The gold star marks the global optimum. Edges follow the min-ORC direction deterministically. The 406 optima collapse into 69 funnels; 91% converge to a top-5% attractor. The circular layout groups funnels; spatial distance does not encode fitness or Hamming distance.

Figure 2 visualizes the OTG for a representative W-model instance. Clusters of optima flow deterministically toward high-quality attractors from local curvature alone, without stochastic trials or perturbation distance choices.

6.4 Convergence Under Sampling

On W-model at $N=16$, subsampling R optima uniformly (20 subsamples $\times$ 10 instances), per-optimum metrics converge fast: escape rate within 2% error at $R=5$, <1% at $R=20$. Structural metrics converge slowly—at 50% coverage, attractor rank is still 3.4$\times$ worse than ground truth, because the OTG cannot route to unsampled attractors. Per-optimum ORC metrics are thus reliable from $\sim$20 restarts at any scale; full OTG structural analysis remains limited to enumerable landscapes.

6.5 Limitations

The strongest OTG results (up to 9.2$\times$ attractor quality improvement over LON-d1) come from full enumeration at $N=16$ on W-model—a scale where exhaustive analysis is feasible but not representative of real-world problem sizes. At $N=100$ on MAX-SAT, Rand-ILS outperforms ORC+Pert in wall-clock time when evaluations are cheap ($\ll$1 ms). On QAP, ORC's advantage over MinGap is marginal (1.06$\times$) on random instances; structured QAPLIB instances may behave differently. For neighborhoods *without* the disjoint-neighborhood property (e.g., 3-opt, LK moves), Proposition 1's decomposition does not apply and ORC requires the full $O(k^3)$ Hungarian algorithm; characterizing ORC's behavior on such graphs is an open question. A full comparison with the flacco/ELA ecosystem [8] in an algorithm selection pipeline remains future work.

7 Conclusion

The ORC Transition Graph demonstrates that optimal transport over neighborhood fitness distributions captures cross-basin structure invisible to 1-hop methods, with the attractor quality gap over LON-d1 reaching up to 9.2$\times$ on epistatic W-model landscapes at $N = 16$. The disjoint-neighborhood property (Corollary 1) gives the framework a unified foundation across the three foundational neighborhoods in combinatorial optimization: bit-flips, permutation swaps, and TSP 2-opt. ORC's per-step escape advantage grows monotonically with neighborhood size (3.1$\times$ to 18.3$\times$ over random on TSP from $n=8$ to $n=50$; analogous trends on binary and swap spaces), though translating this into algorithmic gains remains cost-dependent. OTG structural properties—particularly the acyclic-to-cyclic crossover—correlate with when ORC-guided perturbation adds value in ILS, providing practitioners with a diagnostic signal. Per-optimum metrics converge from $\sim$20 samples, making ORC a practical addition to the FLA toolkit.

Reproducibility. All code and data are available at
`https://github.com/rhordoan/RiemannianSwarmSurgery`.

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